

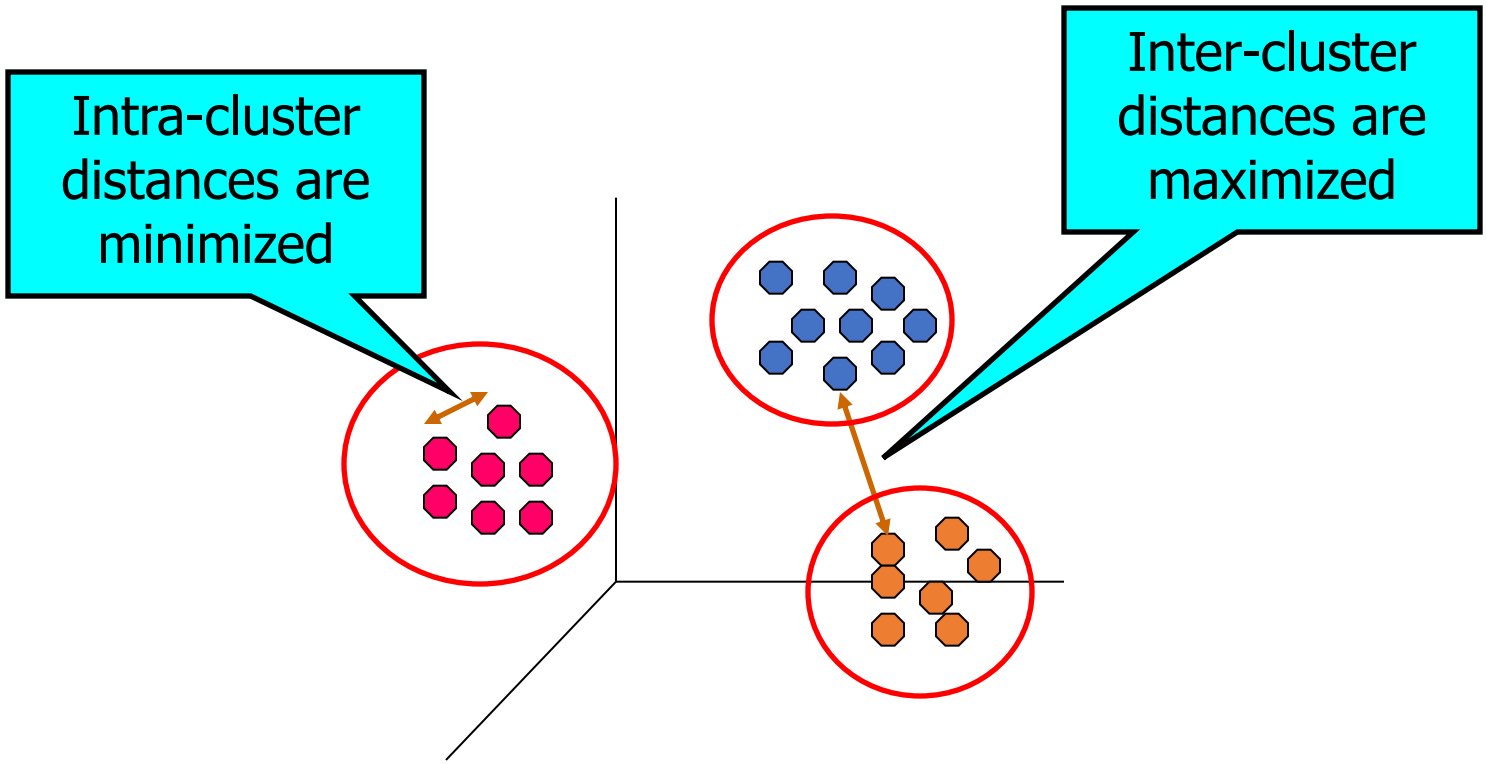
CSE5DMI Data Mining

Week 09
CLUSTER ANALYSIS: BASIC CONCEPTS AND ALGORITHMS

La Trobe University CRICOS Provider Code Number 00115M

What is Cluster Analysis?

- Given a set of objects, place them in groups such that the objects in a group are similar (or related) to one another and different from (or unrelated to) the objects in other groups



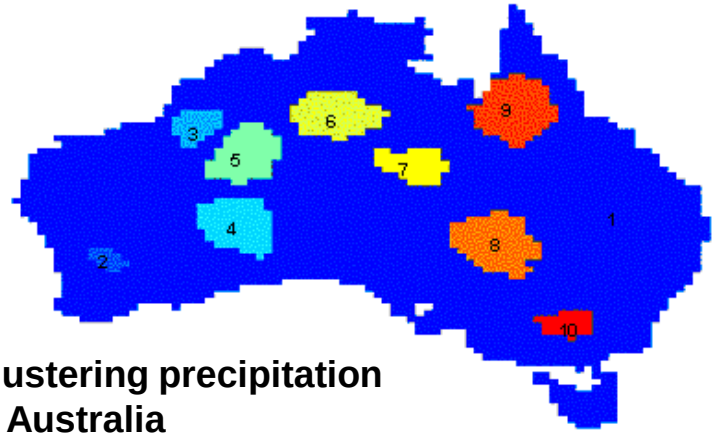
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Applications of Cluster Analysis

- **Understanding**
  - Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations

	Discovered Clusters	*Industry Group*
1	Applied-Matl-DOWN,Bay-Network-Down,3-COM-DOWN, Cabletron-Sys-DOWN,CISCO-DOWN,HP-DOWN, DSC-Comm-DOWN,INTEL-DOWN,LSI-Logic-DOWN, Micron-Tech-DOWN,Texas-Inst-Down,Tellabs-Inc-Down, Natl-Semiconduct-DOWN,Oracl-DOWN,SGI-DOWN, Sun-DOWN	Technology1-DOWN
2	Apple-Comp-DOWN,Autodesk-DOWN,DEC-DOWN, ADV-Micro-Device-DOWN,Andrew-Corp-DOWN, Computer-Assoc-DOWN,Circuit-City-DOWN, Compaq-DOWN, EMC-Corp-DOWN, Gen-Inst-DOWN, Motorola-DOWN,Microsoft-DOWN,Scientific-Atl-DOWN	Technology2-DOWN
3	Fannie-Mae-DOWN,Fed-Home-Loan-DOWN, MBNA-Corp-DOWN,Morgan-Stanley-DOWN	Financial-DOWN
4	Baker-Hughes-UP,Dresser-Inds-UP,Halliburton-HLD-UP, Louisiana-Land-UP,Phillips-Petro-UP,Unocal-UP, Schlumberger-UP	Oil-UP

- **Summarization**
  - Reduce the size of large data sets



Clustering precipitation in Australia

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Notion of a Cluster can be Ambiguous



How many clusters?



Six Clusters



Two Clusters



Four Clusters

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Types of Clusterings

- A clustering is a set of clusters

- Important distinction between hierarchical and partitional sets of clusters

  - Partitional Clustering
  - A division of data objects into non-overlapping subsets (clusters)

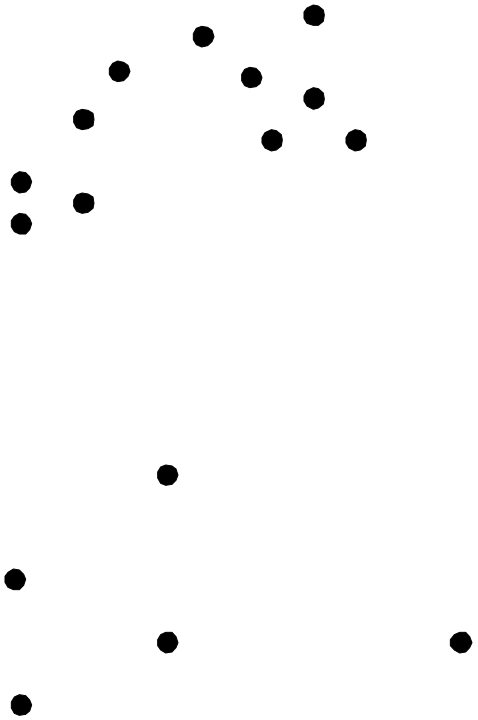
  - Hierarchical clustering
  - A set of nested clusters organized as a hierarchical tree

The most discussed distinction among different types of clustering is whether the set of clusters is nested or unnested, or in more traditional terminology, hierarchical or partitional .

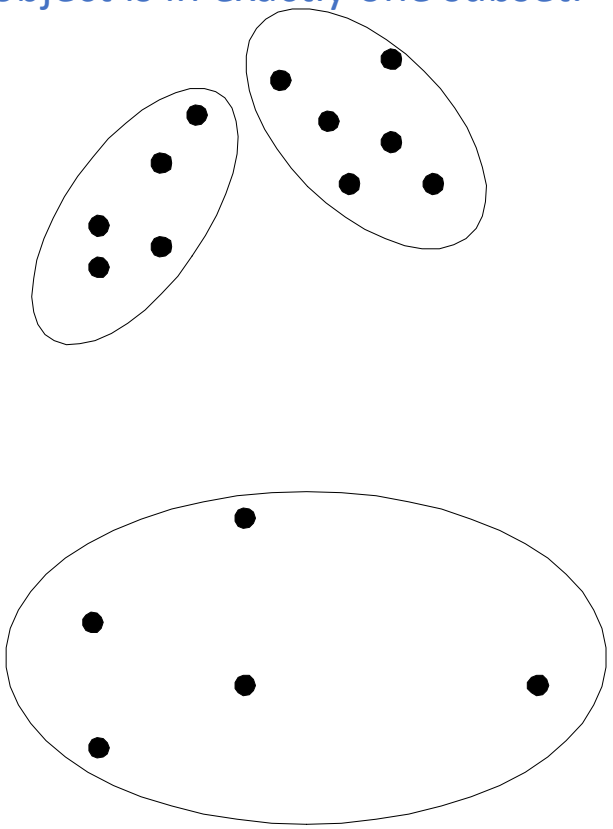
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Partitional Clustering

A division of the set of data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset.



Original Points

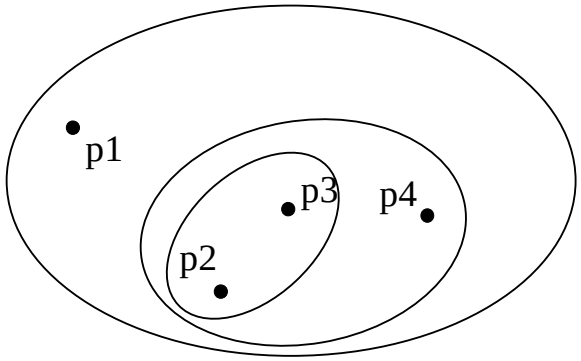


A Partitional Clustering

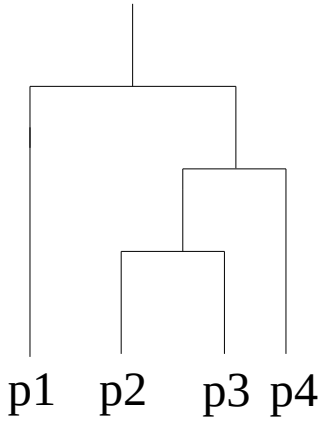
We do the grouping without using class labels.
We just use features or attributes.

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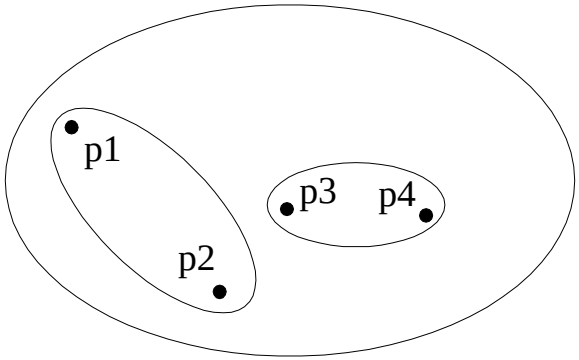
Hierarchical Clustering



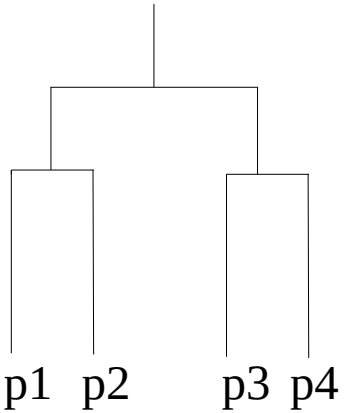
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

If we permit clusters to have subclusters, then we obtain a hierarchical clustering. It would be a set of nested clusters that are organized as a tree. Each node / cluster in the tree is the union of its children, and the root of the tree is the cluster containing all the objects.

Other Distinctions Between Sets of Clusters

- Exclusive versus non-exclusive
  - In non-exclusive clusterings, points may belong to multiple clusters.
    - Can belong to multiple classes or could be ‘border’ points
  - Fuzzy clustering (one type of non-exclusive)
    - In fuzzy clustering, a point belongs to every cluster with some weight between 0 and 1
    - Weights must sum to 1
    - Probabilistic clustering has similar characteristics
    - In practice, a fuzzy clustering is often converted to an exclusive clustering by assigning each object to the cluster in which is membership weight is highest.
- Partial versus complete
  - In some cases, we only want to cluster some of the data

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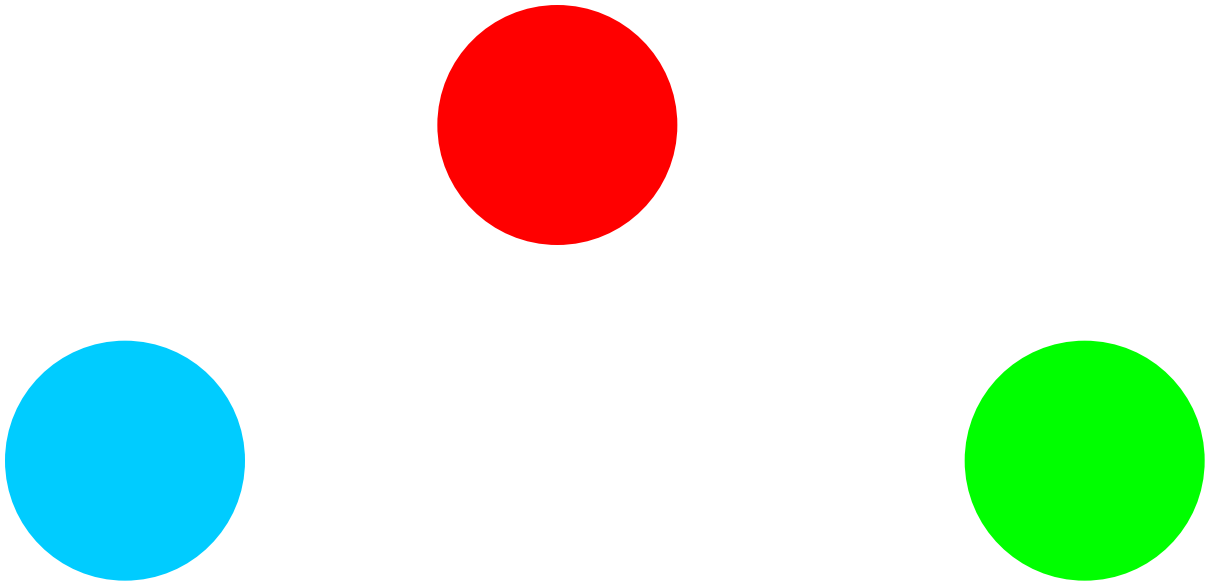
Types of Clusters

- Well-separated clusters
- Prototype-based clusters
- Contiguity-based clusters
- Density-based clusters
- Described by an Objective Function

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Types of Clusters: Well-Separated

- Well-Separated Clusters:
  - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.



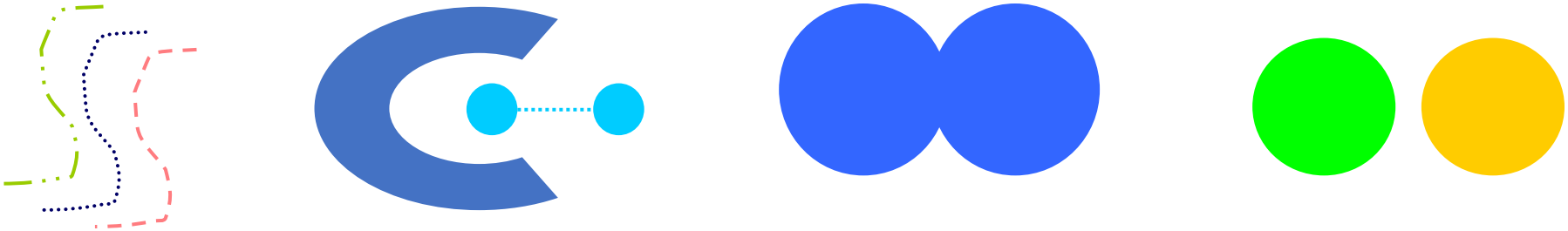
3 well-separated clusters

Note: this is just an example. Well-separated clusters can have any shapes.

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Types of Clusters: Contiguity-Based

- Contiguous Cluster (Nearest neighbor or Transitive)
  - A cluster is a set of points such that a point in a cluster is closer (or more similar) to one or more other points in the cluster than to any point not in the cluster.
  - This definition of a cluster is useful when clusters are irregular or intertwined

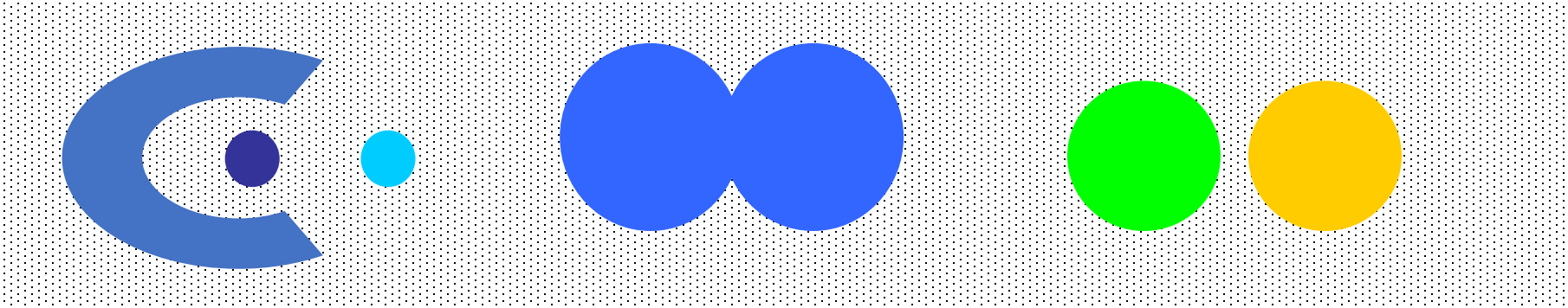


8 contiguous clusters

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Types of Clusters: Density-Based

- Density-based
  - A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
  - Used when the clusters are irregular or intertwined, and when noise and outliers are present.



Here shows some density-based clusters for data created by adding noise to the data in the previous slide.

6 density-based clusters

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Clustering Algorithms

- K-means and its variants
- Hierarchical clustering
- Density-based clustering

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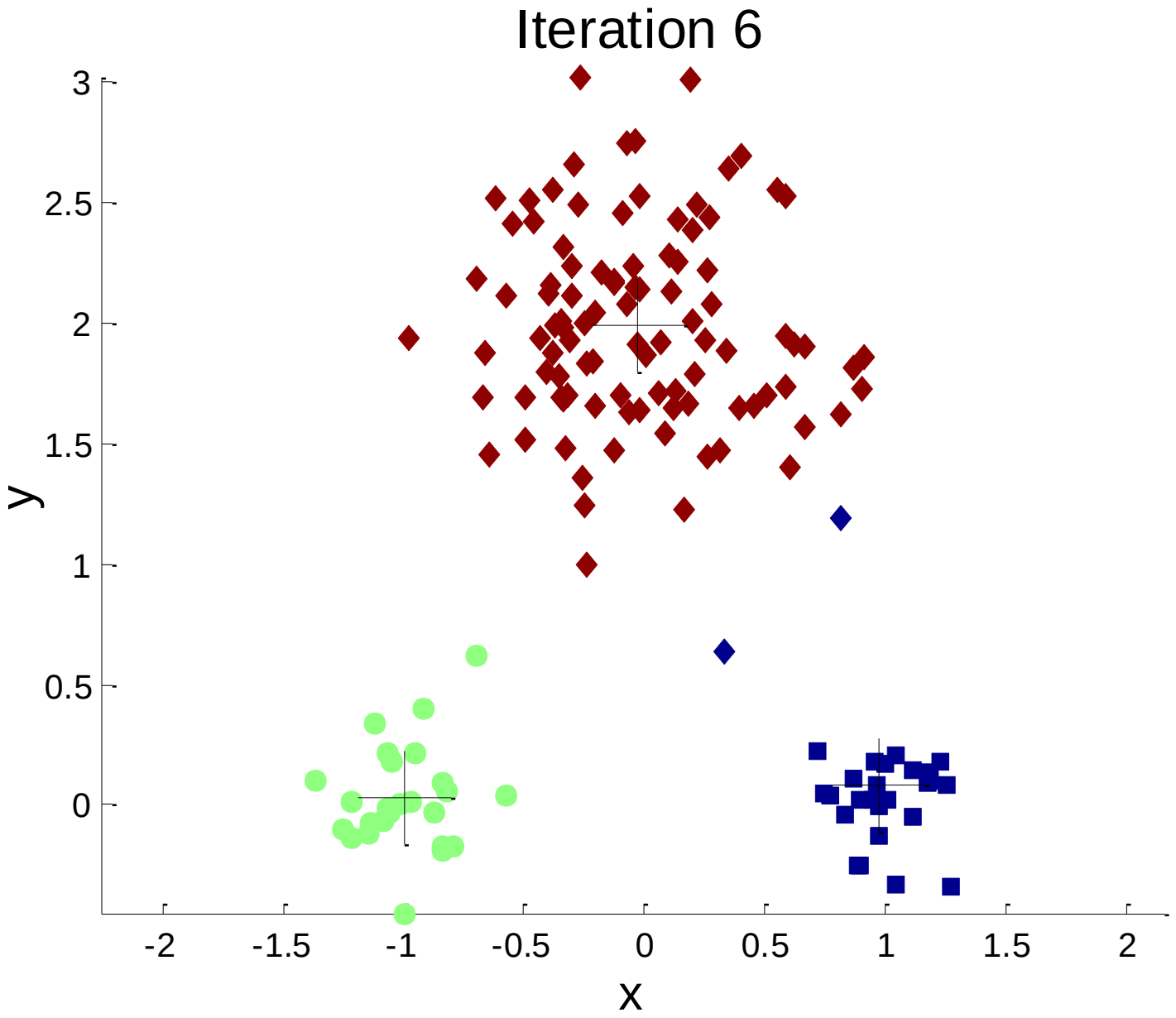
K-means Clustering

- Partitional clustering approach
- Number of clusters, K, must be specified
- Each cluster is associated with a centroid (center point)
- Each point is assigned to the cluster with the closest centroid
- The basic algorithm is very simple

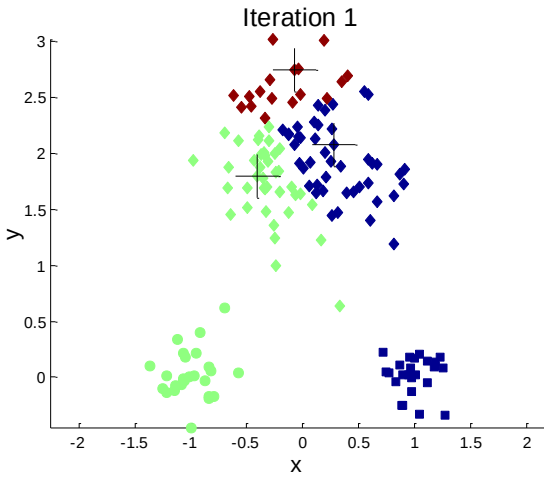
1: Select K points as the initial centroids.
2: **repeat**
3: Form K clusters by assigning all points to the closest centroid.
4: Recompute the centroid of each cluster.
5: **until** The centroids don't change

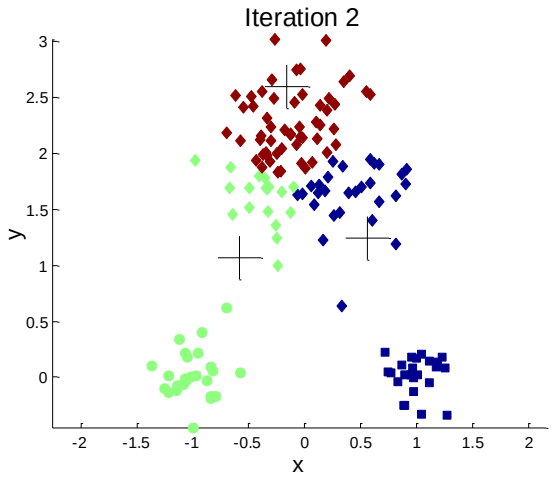
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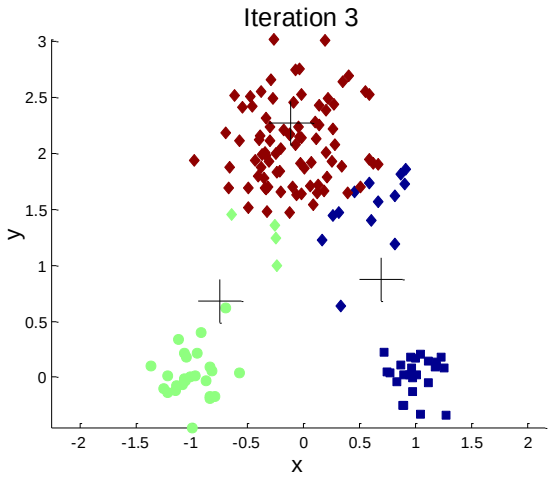
Example of K-means Clustering

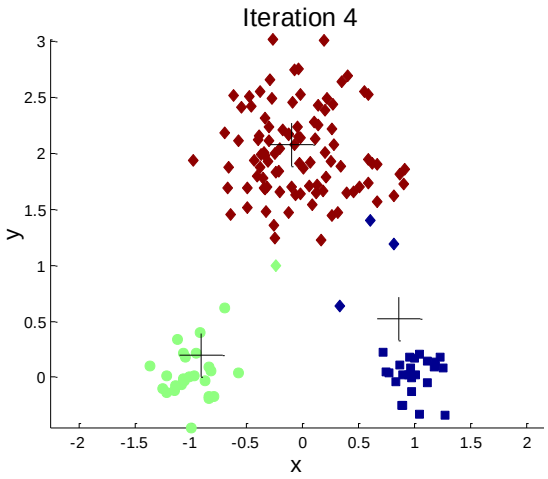


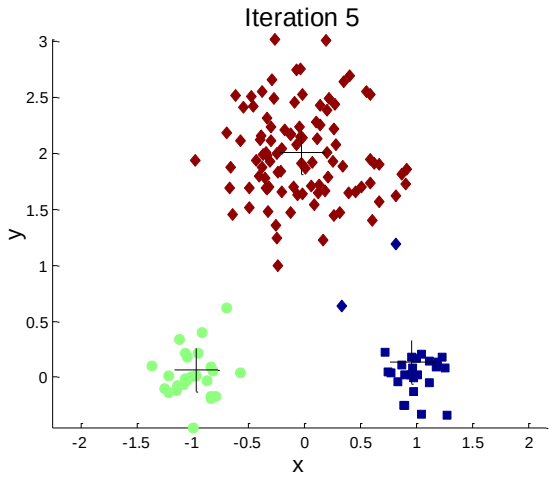
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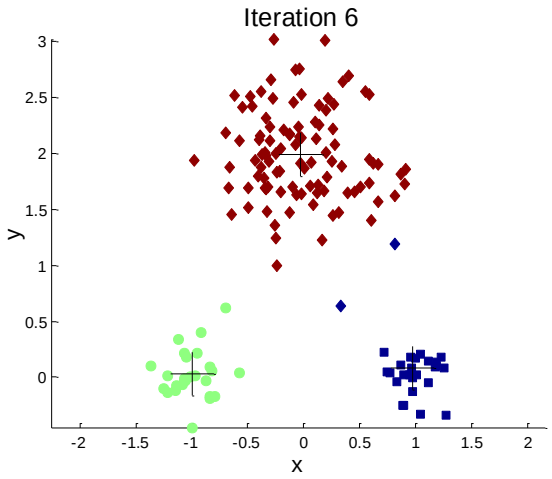












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K-means Clustering – Details

- Simple iterative algorithm.
  - Choose initial centroids;
  - repeat {assign each point to a nearest centroid; re-compute cluster centroids}
  - until centroids stop changing.
- Initial centroids are often chosen randomly.
  - Clusters produced can vary from one run to another
- The centroid is (typically) the mean of the points in the cluster, but other definitions are possible.
- K-means will converge for common proximity measures with appropriately defined centroid
- Most of the convergence happens in the first few iterations.
  - Often the stopping condition is changed to 'Until relatively few points change clusters'
- Complexity is O(n * K * I * d)
  - n = number of points, K = number of clusters,
    I = number of iterations, d = number of attributes

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K-means Objective Function

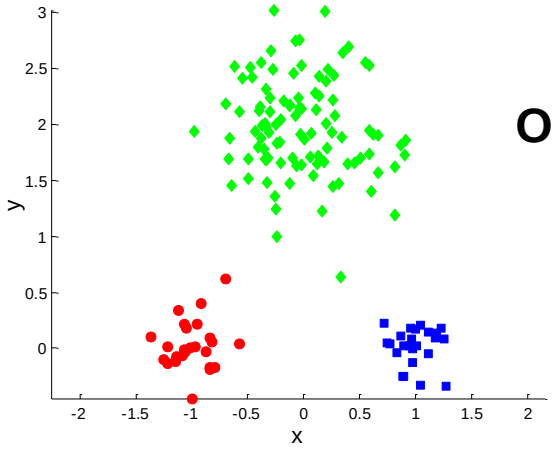
- A common objective function (used with Euclidean distance measure) is Sum of Squared Error (SSE)
  - For each point, the error is the distance to the nearest cluster center
  - To get SSE, we square these errors and sum them.

$$SSE = \sum_{i=1}^{K} \sum_{x \in C_i} dist^2(m_i, x)$$

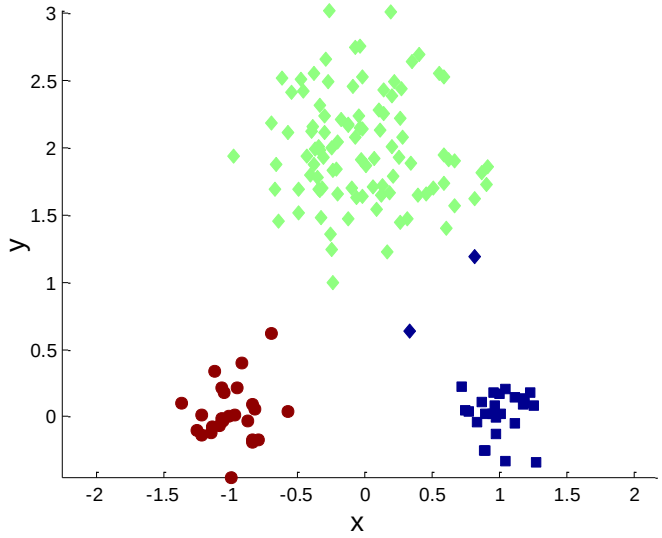
  - x is a data point in cluster C_i and m_i is the centroid (mean) for cluster C_i
  - SSE improves in each iteration of K-means until it reaches a local or global minima.
  - Given two clusters, we can choose the one with the smallest error
  - One easy way to reduce SSE is to increase K, the number of clusters
    - A good clustering with smaller K can have a lower SSE than a poor clustering with higher K

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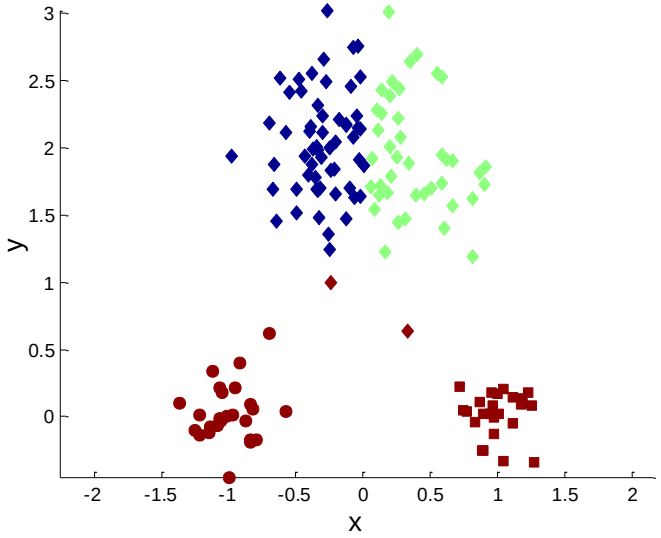
Two different K-means Clusterings



Original Points



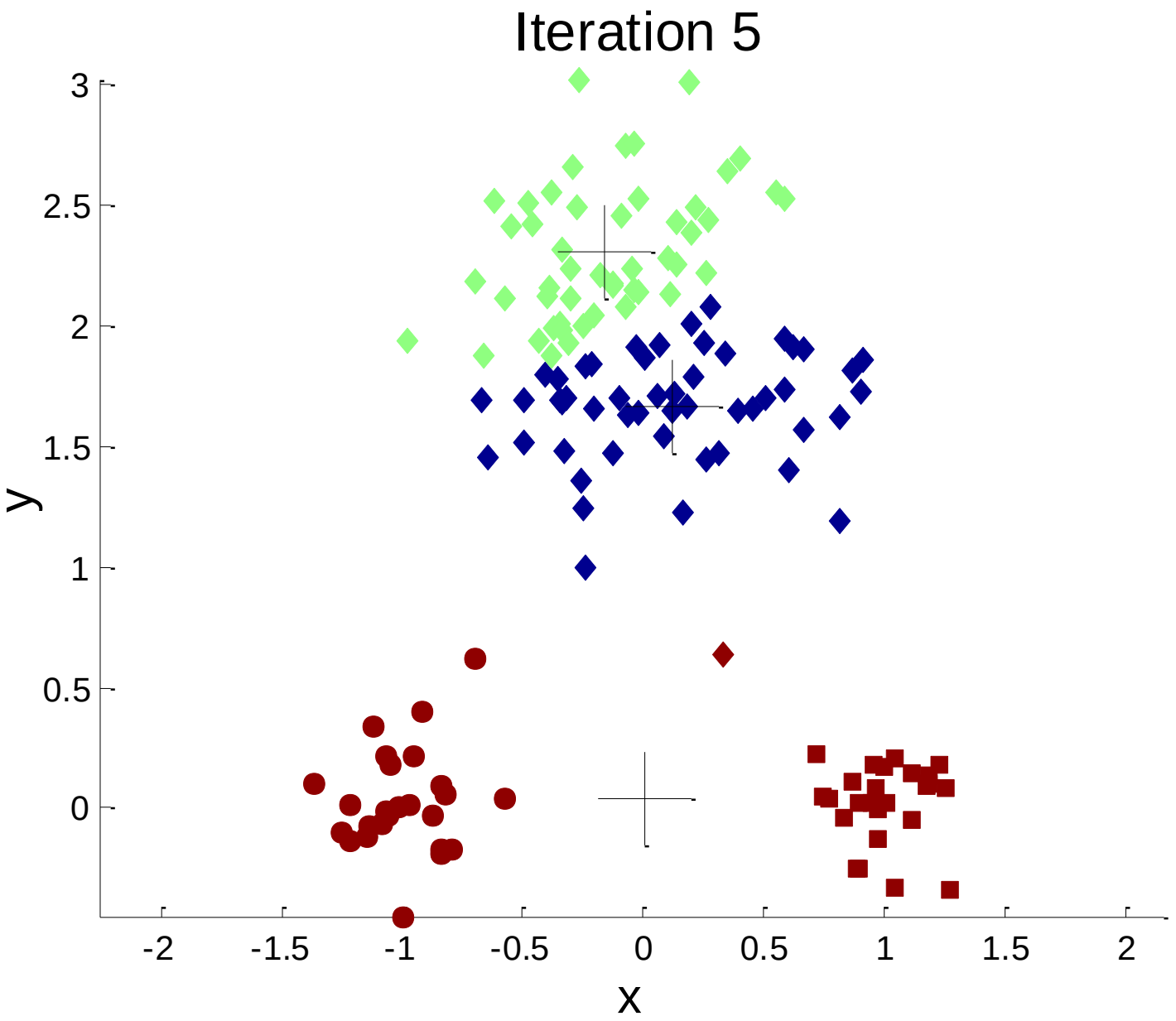
Optimal Clustering



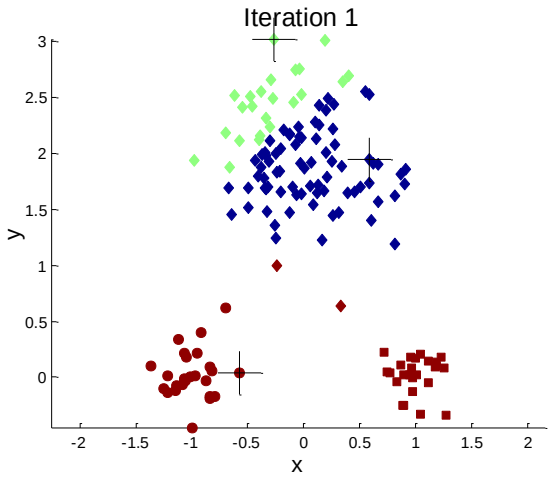
Sub-optimal Clustering

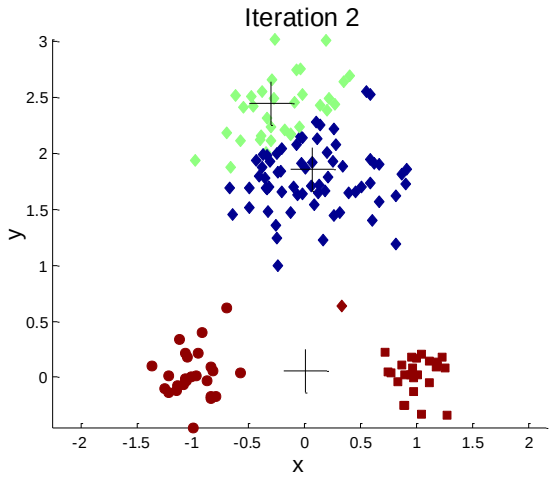
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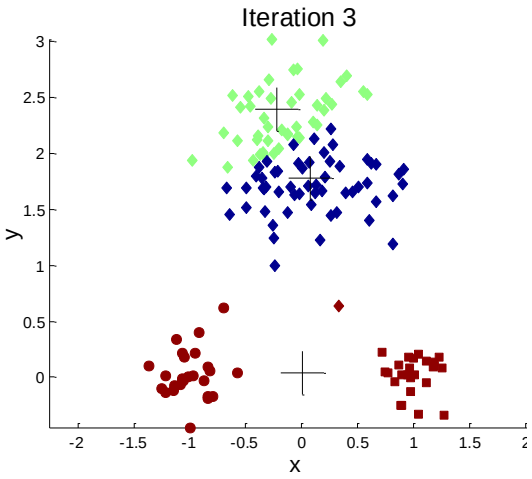
Importance of Choosing Initial Centroids …

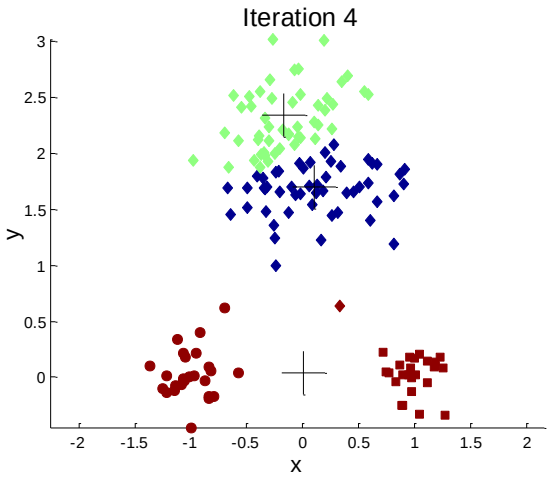


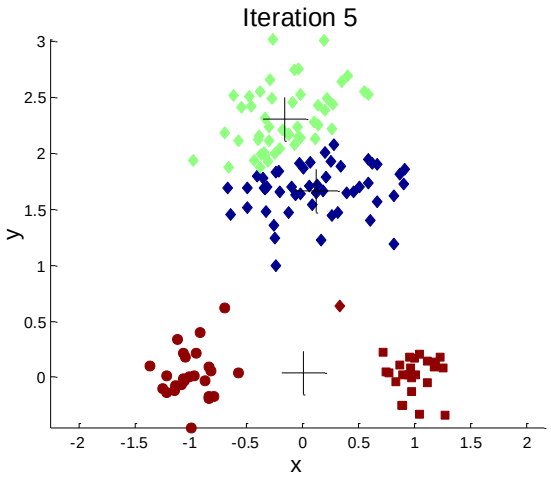
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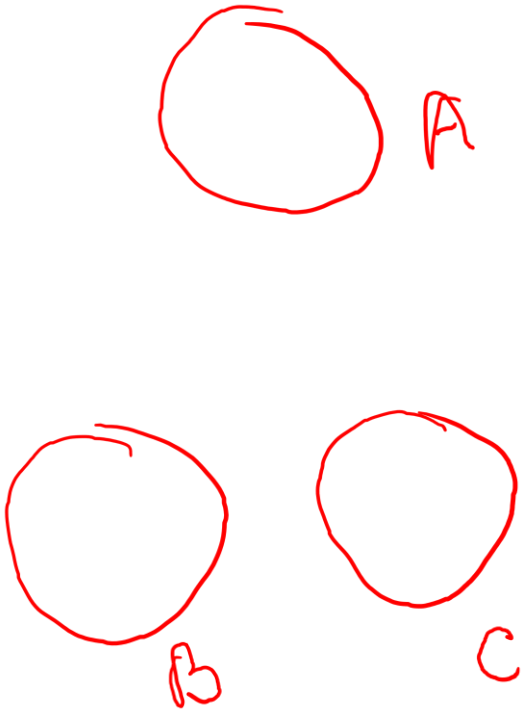




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Importance of Choosing Initial Centroids

- Depending on the choice of initial centroids, B and C may get merged or remain separate



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Problems with Selecting Initial Points

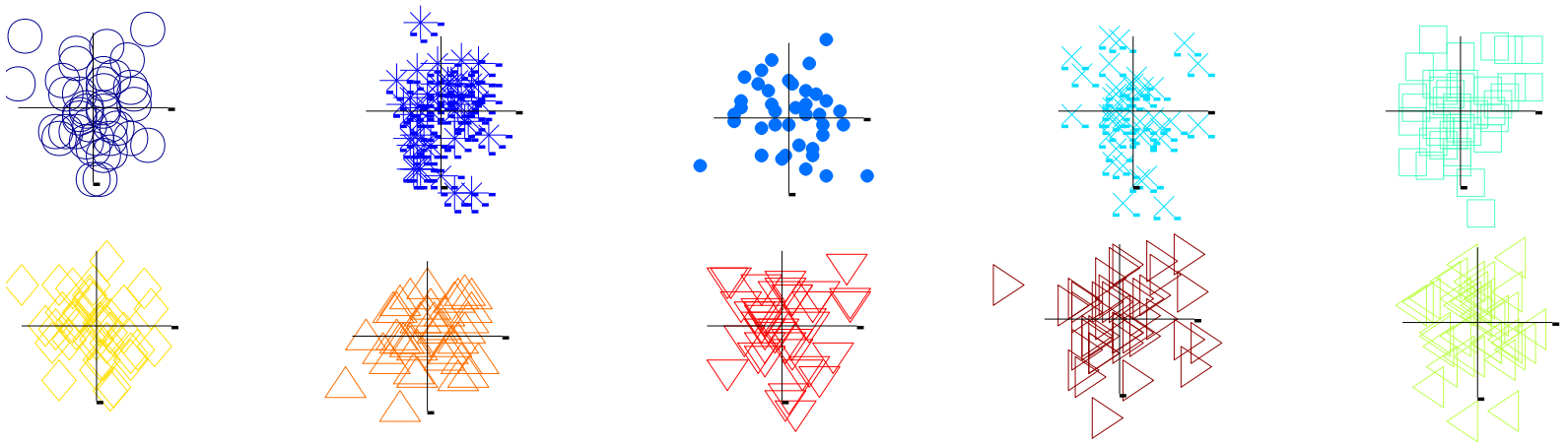
- If there are K 'real' clusters then the chance of selecting one centroid from each cluster is small.
  - Chance is relatively small when K is large
  - If clusters are the same size, n, then

$$P = \frac{\text{number of ways to select one centroid from each cluster}}{\text{number of ways to select } K \text{ centroids}} = \frac{K!n^K}{(Kn)^K} = \frac{K!}{K^K}$$

  - For example, if K = 10, then probability = $10!/10^{10} = 0.00036$
  - Sometimes the initial centroids will readjust themselves in 'right' way, and sometimes they don't
  - Consider an example of five pairs of clusters

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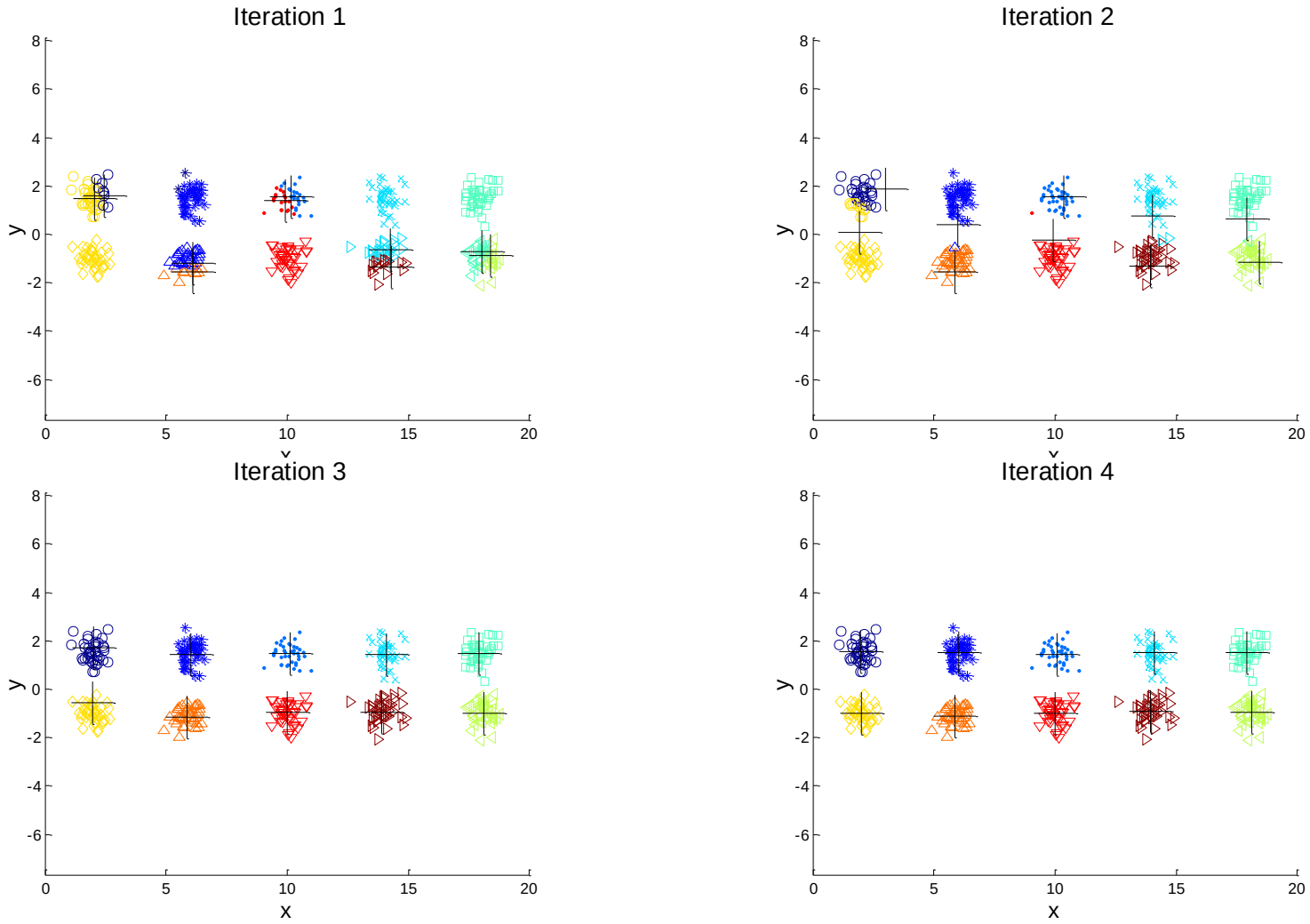
10 Clusters Example



Starting with two initial centroids in one cluster of each pair of clusters

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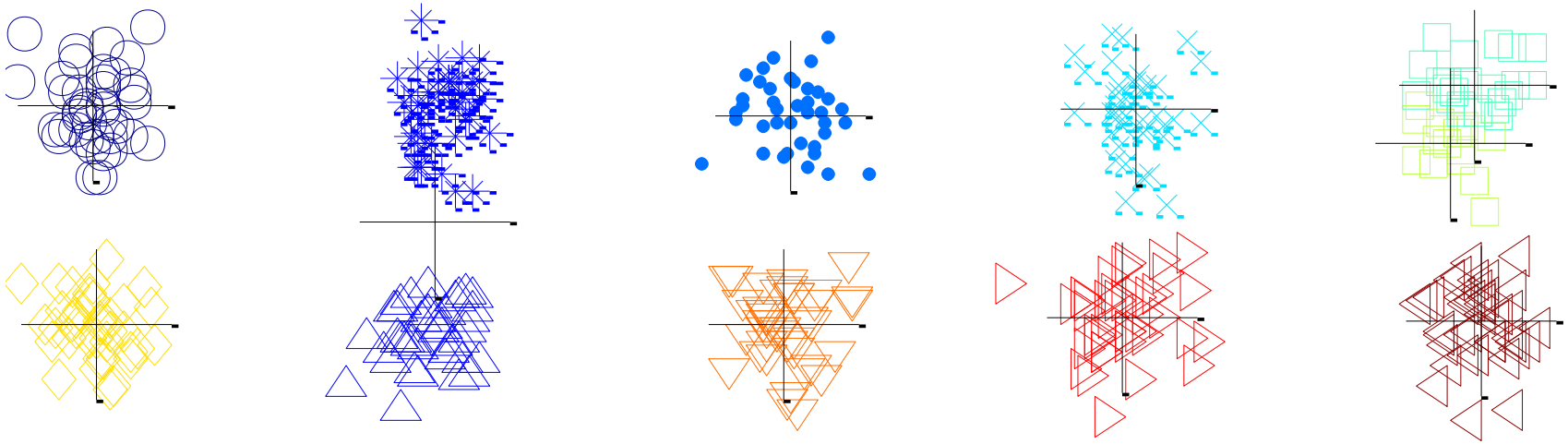
10 Clusters Example



Starting with two initial centroids in one cluster of each pair of clusters

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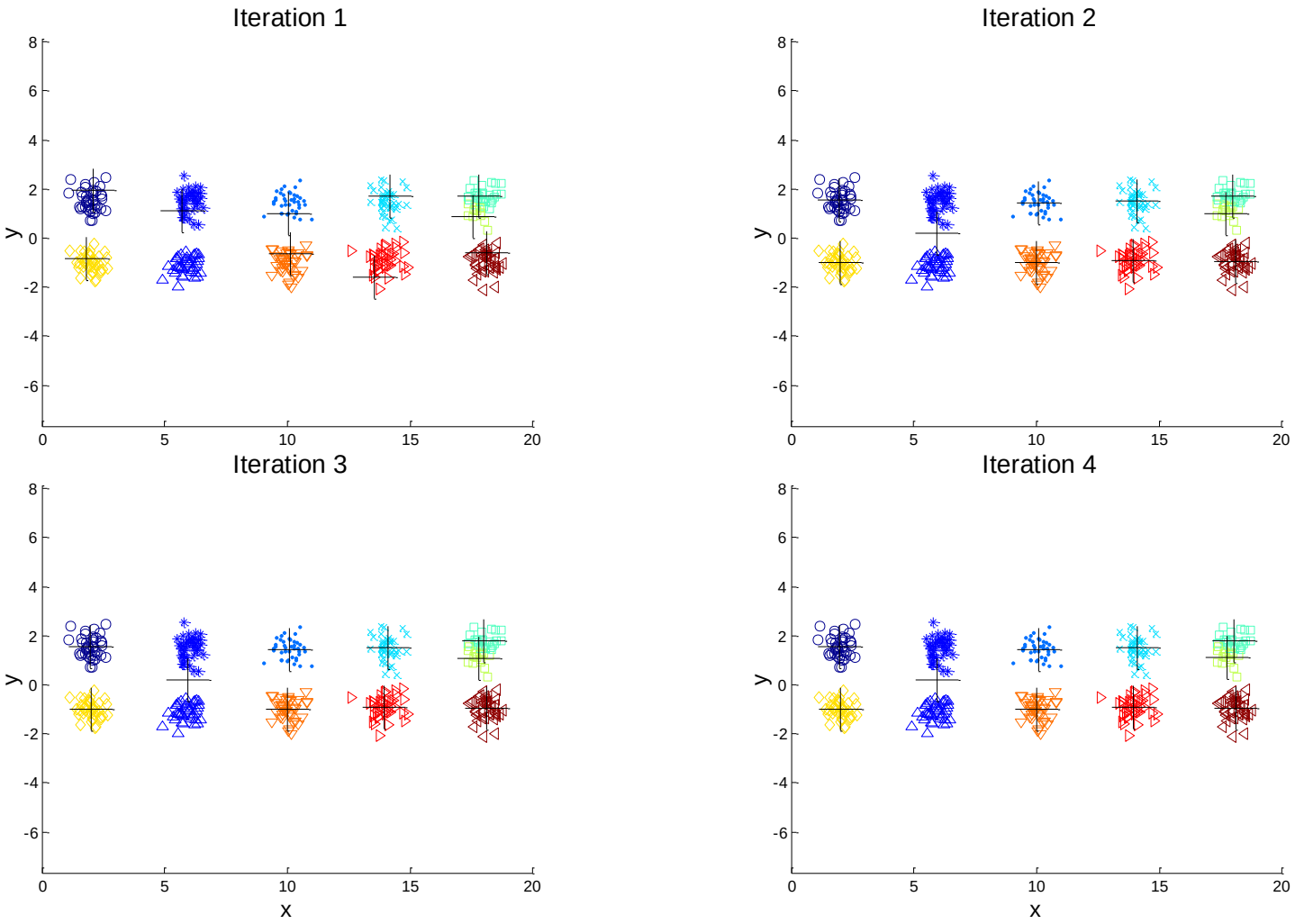
10 Clusters Example



Starting with some pairs of clusters having three initial centroids, while other have only one.

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10 Clusters Example



Starting with some pairs of clusters having three initial centroids, while other have only one.

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Solutions to Initial Centroids Problem

- Multiple runs
  - Helps, but probability is not on your side
- Use some strategy to select the k initial centroids and then select among these initial centroids
  - Select most widely separated
    - K-means++ is a robust way of doing this selection
  - Use hierarchical clustering to determine initial centroids
- Bisecting K-means
  - Not as susceptible to initialization issues

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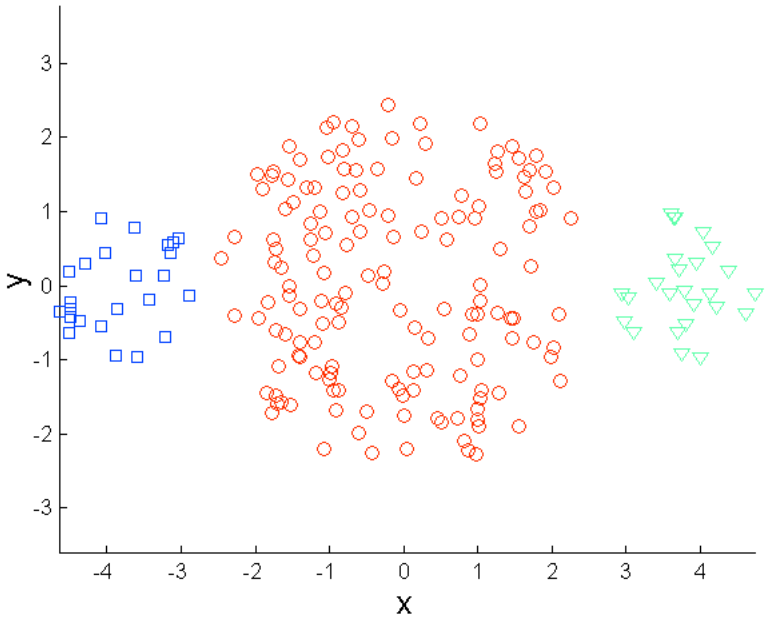
Limitations of K-means

- K-means has problems when clusters are of differing
  - Sizes
  - Densities
  - Non-globular shapes

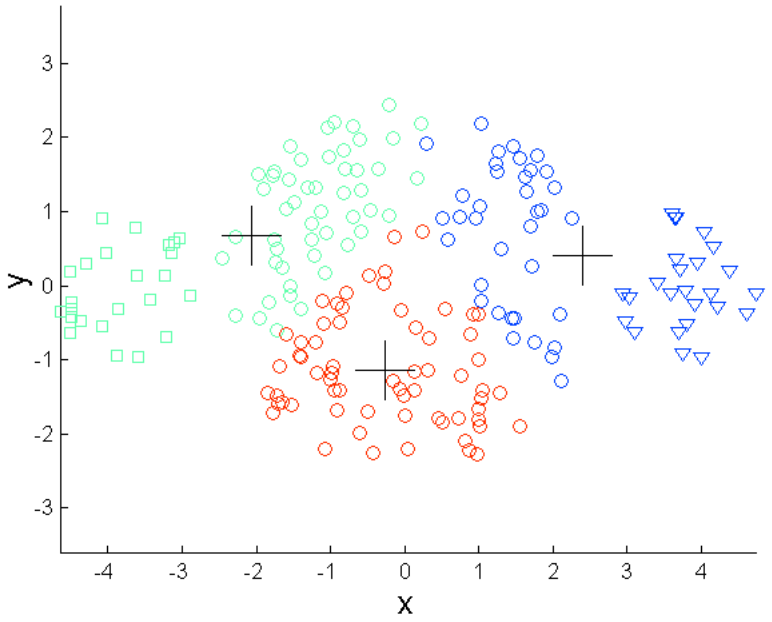
- K-means has problems when the data contains outliers.
  - One possible solution is to remove outliers before clustering

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Limitations of K-means: Differing Sizes



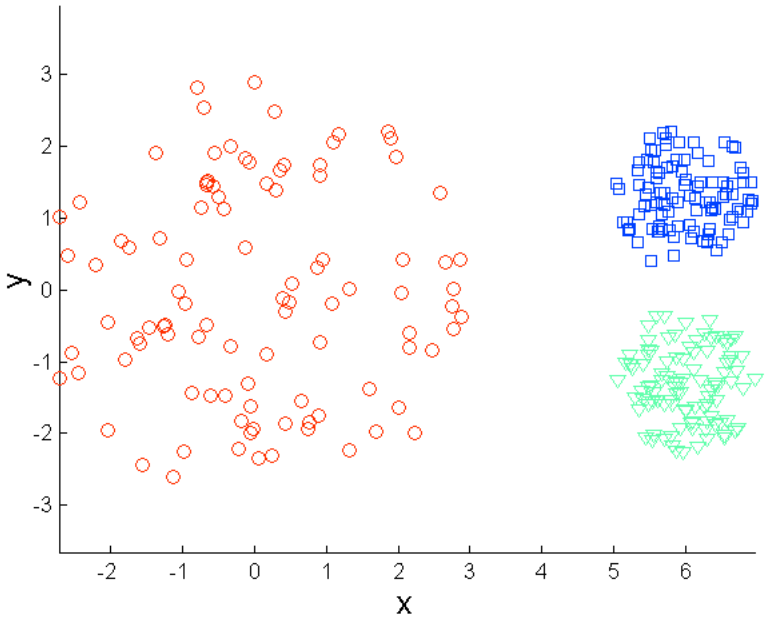
Original Points



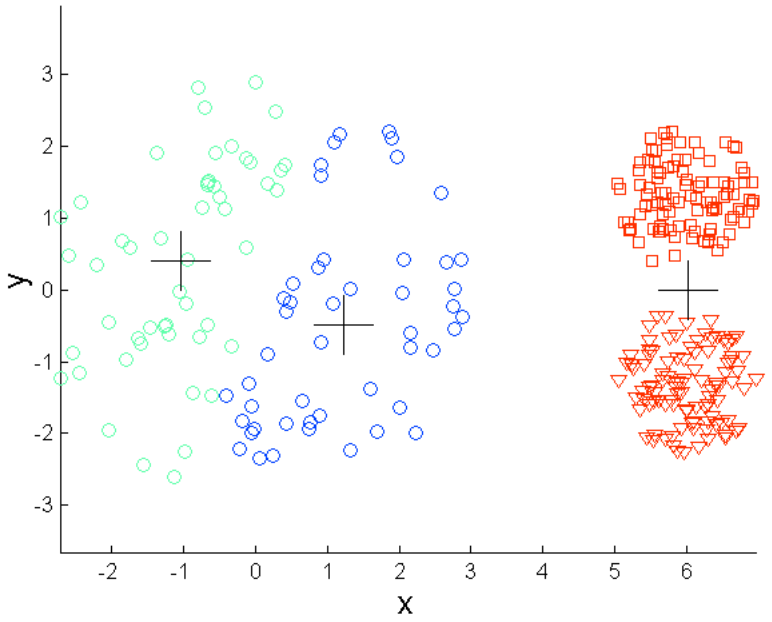
K-means (3 Clusters)

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Limitations of K-means: Differing Density



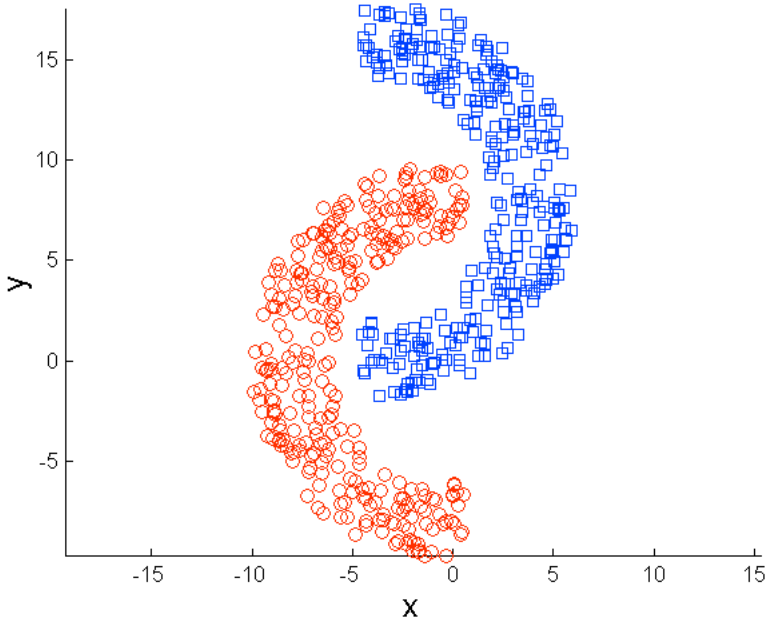
Original Points



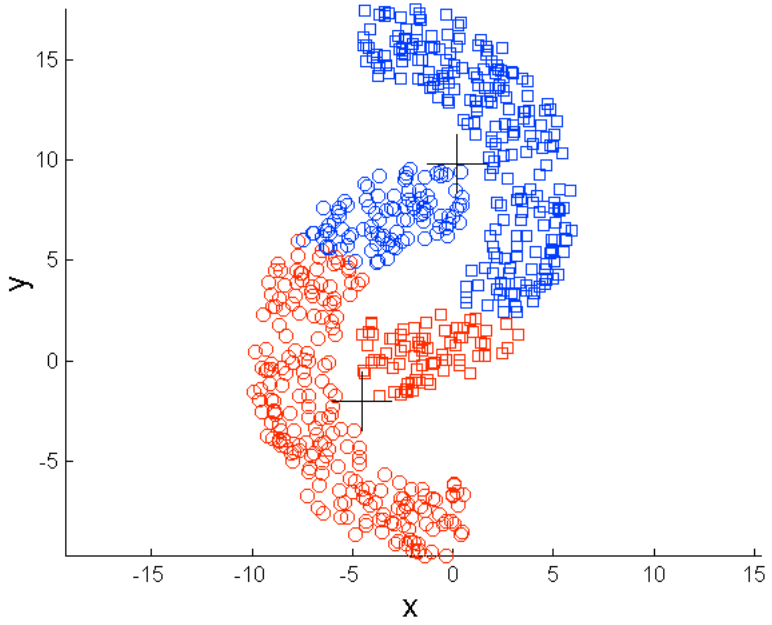
K-means (3 Clusters)

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Limitations of K-means: Non-globular Shapes



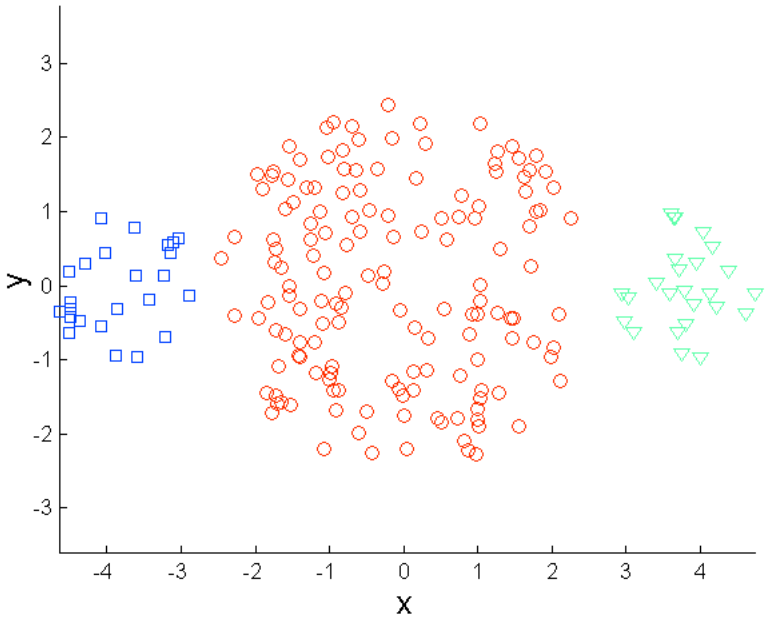
Original Points



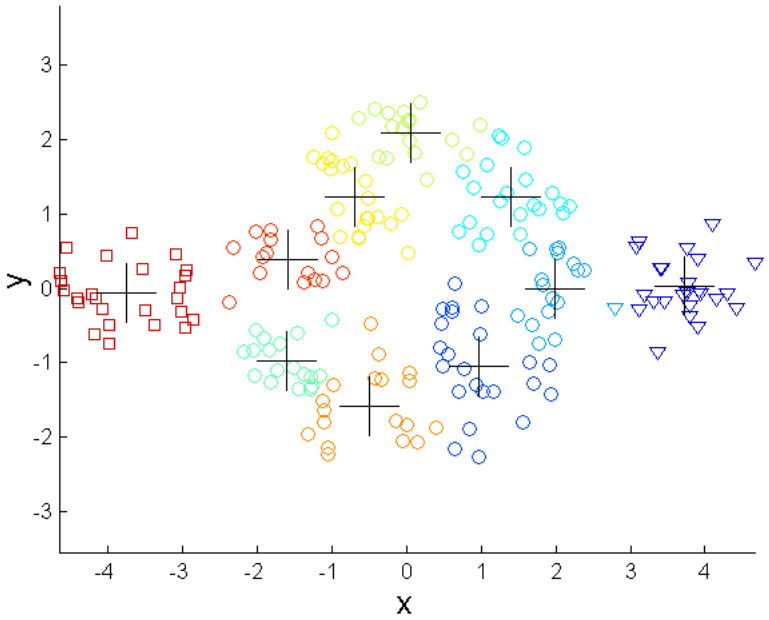
K-means (2 Clusters)

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Overcoming K-means Limitations



Original Points

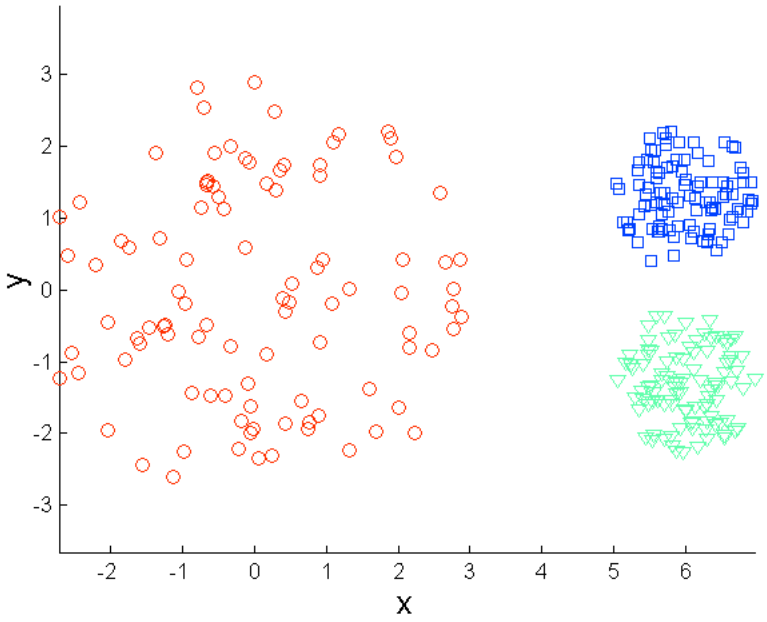


K-means Clusters

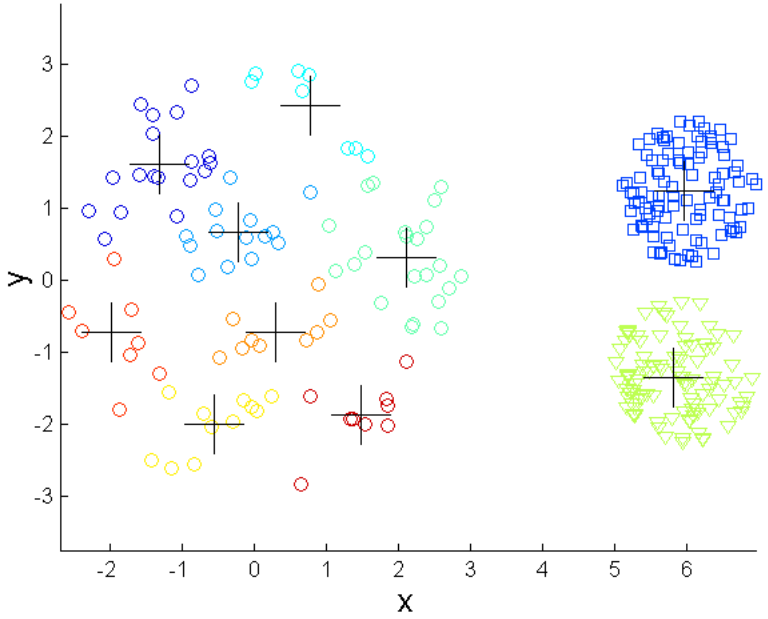
One solution is to find a large number of clusters such that each of them represents a part of a natural cluster. But these small clusters need to be put together in a post-processing step.

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Overcoming K-means Limitations



Original Points

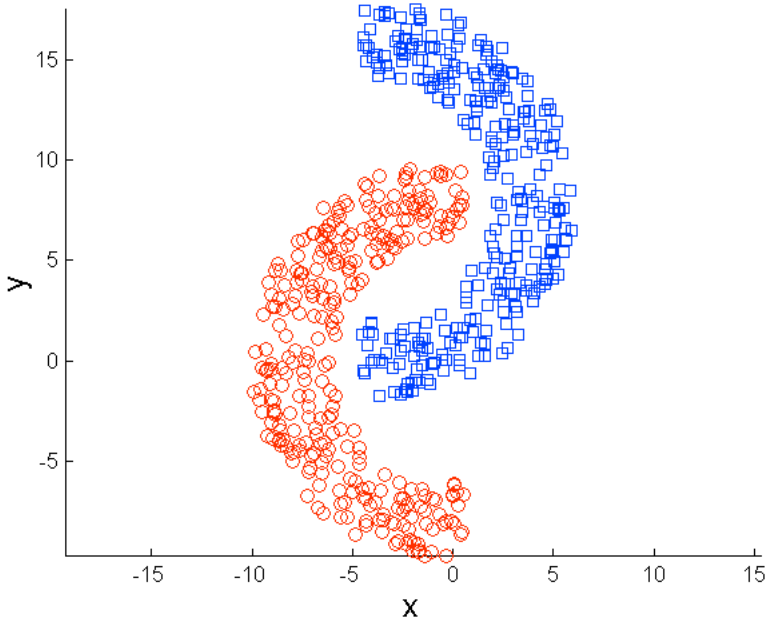


K-means Clusters

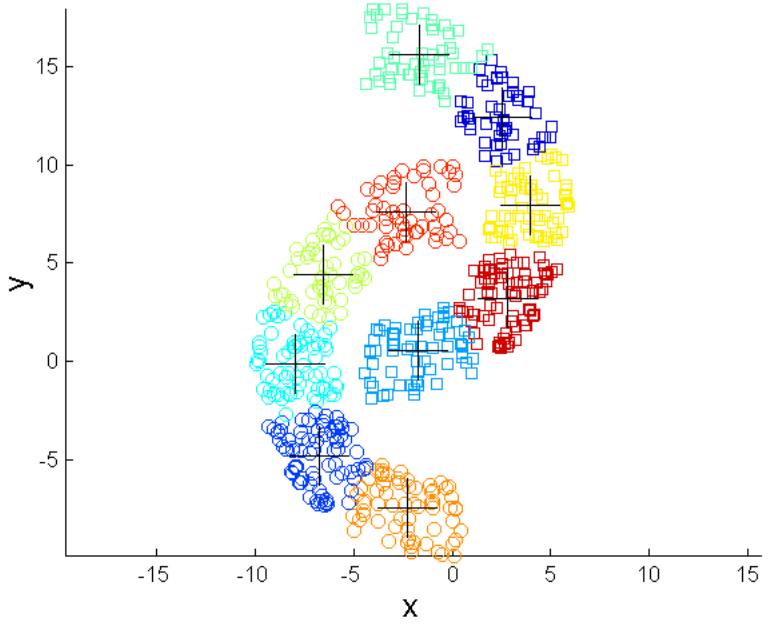
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Overcoming K-means Limitations



Original Points



K-means Clusters

One solution is to find a large number of clusters such that each of them represents a part of a natural cluster. But these small clusters need to be put together in a post-processing step.

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Density Based Clustering

- Clusters are regions of high density that are separated from one another by regions on low density.



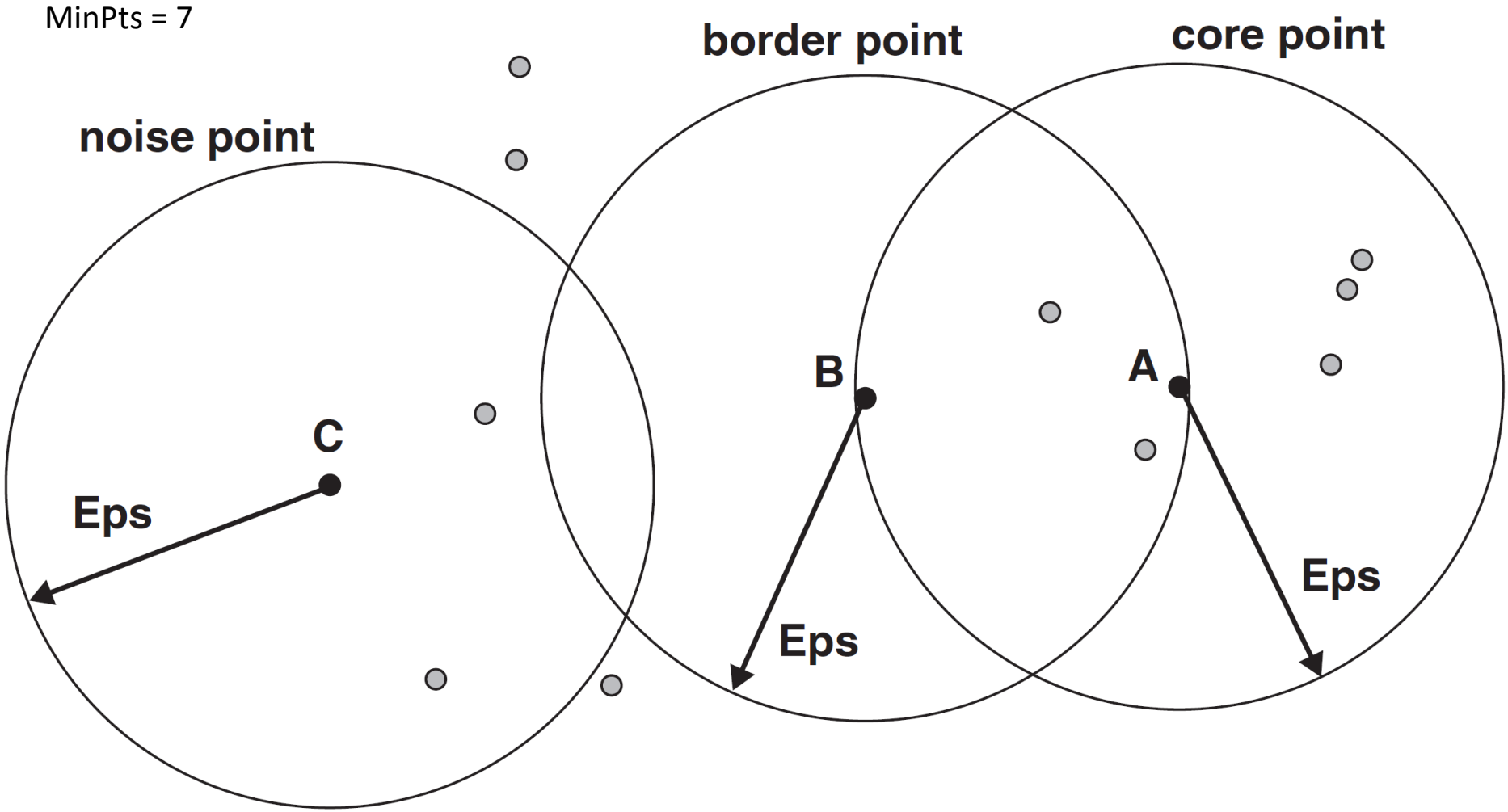
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DBSCAN

- DBSCAN is a density-based algorithm.
  - Density = number of points within a specified radius (Eps)
  - A point is a core point if it has at least a specified number of points (MinPts) within Eps
    - These are points that are at the interior of a cluster
    - Counts the point itself
  - A border point is not a core point, but is in the neighborhood of a core point
  - A noise point is any point that is not a core point or a border point

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DBSCAN: Core, Border, and Noise Points

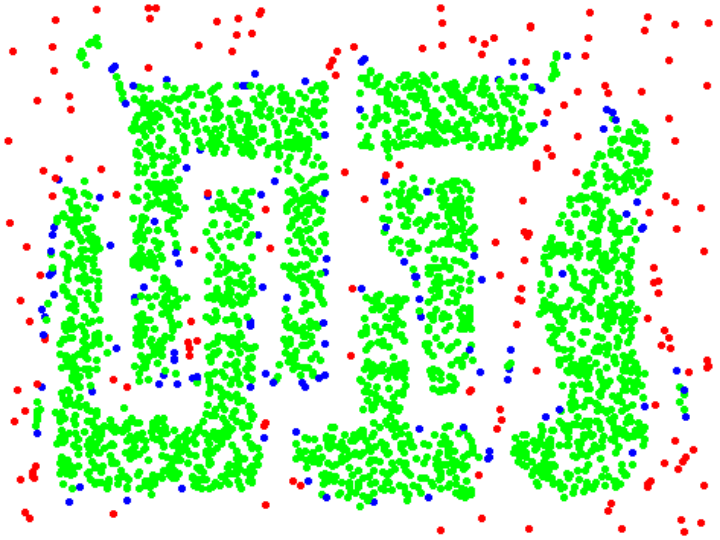


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DBSCAN: Core, Border and Noise Points



Original Points



Point types: core, border and noise

Eps = 10, MinPts = 4

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DBSCAN Algorithm

- Form clusters using core points, and assign border points to one of its neighboring clusters

1: Label all points as core, border, or noise points.

2: Eliminate noise points.

3: Put an edge between all core points within a distance *Eps* of each other.

4: Make each group of connected core points into a separate cluster.

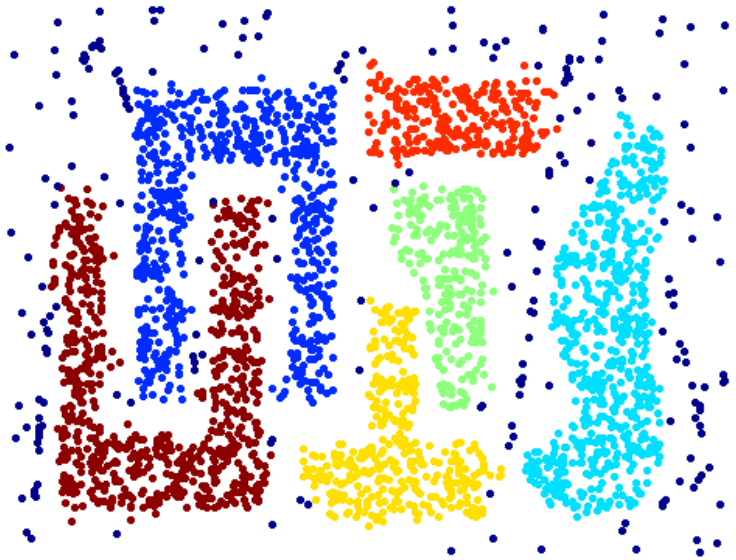
5: Assign each border point to one of the clusters of its associated core points

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When DBSCAN Works Well



Original Points

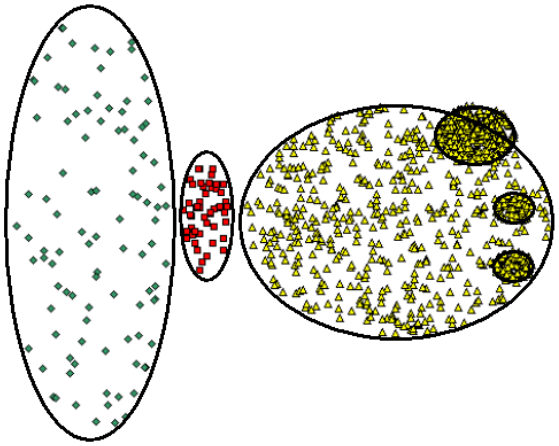


Clusters (dark blue points indicate noise)

- Can handle clusters of different shapes and sizes
- Resistant to noise

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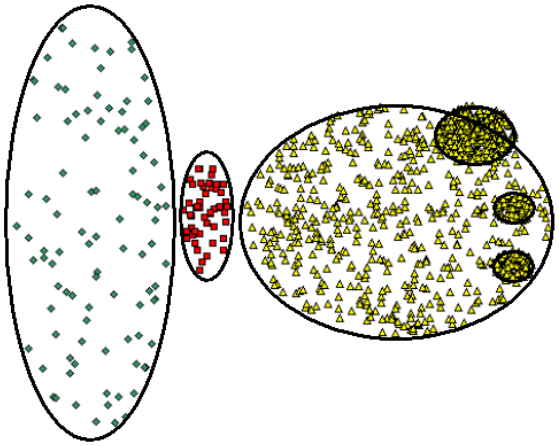
When DBSCAN Does NOT Work Well



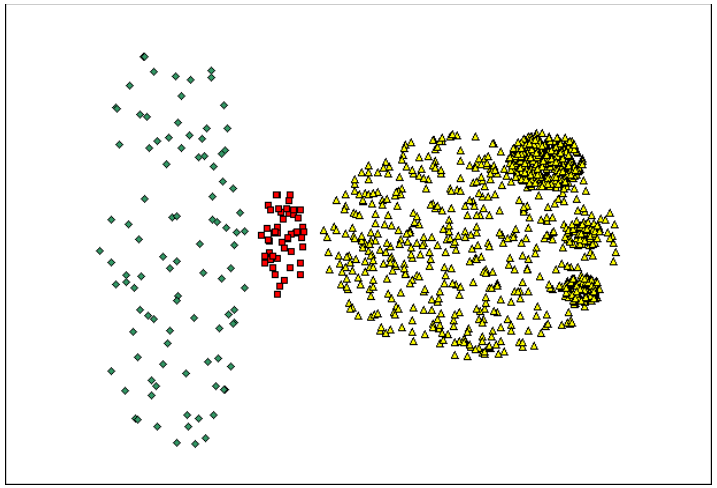
Original Points

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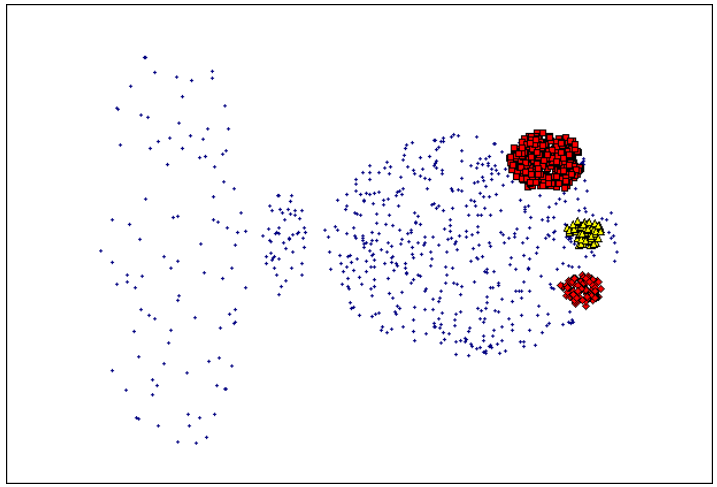
When DBSCAN Does NOT Work Well



Original Points



(MinPts=4, Eps=9.92).



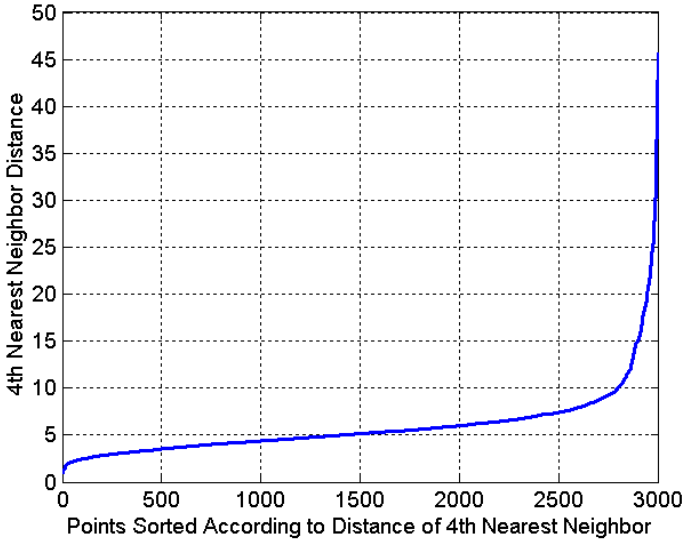
(MinPts=4, Eps=9.75)

- Varying densities
- High-dimensional data

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DBSCAN: Determining EPS and MinPts

- Idea is that for points in a cluster, their k^{th} nearest neighbors are at close distance
- Noise points have the kth nearest neighbor at farther distance
- So, plot sorted distance of every point to its kth nearest neighbor



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Summary

K-Means & DBSCAN

- How does the algorithm form clusters
- When does the algorithm (not) work well

Next week:

- Hierarchical clustering

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Thank you

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